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Crankshaft Reuse Guidelines

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Introduction

This bulletin provides specific visual inspection criteria and reuse guidelines for crankshafts on specific Cummins® engines listed below.

- ISX CM570
- ISX CM870
- ISX CM871
- ISX CM871 E
- ISX12 CM2350 X102
- ISX12 G CM2180 EJ
- ISX12/ISX11.9 CM2250
- ISX15 CM2250
- ISX15 CM2250 SN
- ISX15 CM2350 X101
- PowerGen QSX15 CM2250
- PowerGen QSX15 CM2250 ECF
- QSX CM570
- QSX11.9 CM2250 ECF
- QSX15 CM2250 ECF
- QSX15 CM2250 X115
- QSX15 CM2350 X105
- QSX15 CM2350 X106
- X15 CM2350 X114B
- X15 CM2350 X116B

This bulletin is intended for:

- Progressive damage inspections following a debris-generating malfunction causing contamination of the lubricating oil system.
- Providing high quality, low repair cost guidance for reuse during progressive damage inspections.
- Crankshafts that are installed in the cylinder block, and the engine is installed in the equipment chassis.

Photographs within this bulletin often depict a crankshaft removed from the cylinder block for optimal clarity; however, repair locations are encouraged to perform crankshaft inspection while installed in the engine block and equipment chassis.

Note : If evaluating a crankshaft for reuse during an engine rebuild or overhaul, or the engine or crankshaft requires complete removal from the chassis to inspect, follow the more comprehensive crankshaft inspection and reuse criteria within Procedure 001-016 in Section 01. See the corresponding Service Manual.

Diagnostic Procedures

Similar to the diagnosis of an engine bearing failure, the best practice is to understand as much as possible about the typical operation and service history of the engine during progressive damage investigations, particularly following debris generating failures within the lubricating oil system. To effectively evaluate a crankshaft for reuse, the condition of the bearings needs to be understood. Bearing analysis will reveal a significant amount of information about engine operation, maintenance, and any sources of external debris. A comprehensive guide to engine bearing analysis can be found in Service Bulletin 3810387, Analysis and Prevention of Bearing Failures. Crankshaft reuse criteria can be independent of the bearing condition. Bearings can be deemed beyond reuse while the crankshaft journals show minimal or no damage. However, understanding the condition and severity of damage to the bearings will aid with identification of potential crankshaft damage categories.

Historically, crankshaft journal inspection criteria stated that surface imperfections identified by 'catching a fingernail' do not meet reuse. Although many of the descriptions below still advise using a fingernail to detect the presence of damage, the result of this method does not necessarily imply the crankshaft is damaged beyond reuse. Repairs for this type of imperfection ranged from light polishing to complete crankshaft or engine replacement. Analysis and testing of these components have revealed that all tested components meeting the reuse guidelines outlined in this document would have continued to operate reliably for the duration of engine life. The results of these inspections are documented below into the various categories of damage. The design of the lubricating oil supply, bearing, and crankshaft system is such that a certain level of crankshaft damage can be tolerated. The visual guidelines and descriptions below should be used to make a high quality, cost effective reuse decision.

Definitions

Scoring

- Scoring is the most common type of damage. Surface scratches appearing radially around the circumference of the crankshaft journal. Typically, this condition is indicative of foreign debris in lubricating oil.

Frosted Bands

- High-density microdenting, appearing frosted (white) in appearance. Frosted bands are typically **not** detectable with a fingernail. Crankshaft journal frosting could be a result of stray electrical current through the engine (faulty ground in the equipment's electrical system or an improperly grounded electrical component on the engine).

Varnishing

- Surface film of oxidized oil. Brown or blue in appearance. Varnishing is typically **not** detectable with a fingernail. Moderate varnish build-up on the crankshaft journal is an indication of high-temperature operation and/or extended oil drain intervals.

Raised Burrs

- Displaced crankshaft journal material. Typically this is a result large hard particles in the lubricating oil. This damage is similar to scoring but leaves a raised burr along the length of, or at the termination point of the score tracer. The images in Figure 6 highlight a raised burr. This damage is detectable with a fingernail, and in nearly all cases the crankshaft will **not** be reusable.

Heat Distress (Bluing)

- Heat distress differs from varnishing in the cause, however often looks similar in appearance. Heat distress is discernable from varnishing and typically affects much larger areas of the crankshaft journal and throw. A blue hue will be apparent even throughout the raw forged surface, however, **not** all bluing is detrimental. Bluing is a byproduct of induction hardening of the crankshaft journals and fillet radii. Often part of the crankshaft throw has a blue hue indicating the extent of the hardening process and is **not** detrimental to the performance of the crankshaft. Comparison to crankshaft journals where no bearing damage occurred can help identify the normal blue hue from induction hardening from the blue hue resulting from heat generation of bearing failure.

Inspection and Repair Instructions

WARNING

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

WARNING

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Cleaning the crankshaft journals while installed in the cylinder block can be difficult, but **must** be completed to make an accurate assessment for reuse. Use a lint free cloth/paper and contact cleaner/safety solvent to clean the journals when the bearing shells are removed. If the connecting rods have been removed and the main bearings are being inspected or replaced, spray cleaner into the oil drilling in the rod journal to flush any residual oil and dirt into and out of the oil drilling in the main bearing journal. Rotation of the crankshaft may be necessary to make sure the contact cleaner has an adequate drain path. Use clean compressed air to blow out the drilling.

Basic inspection sequence following a debris-generating component malfunction begins by removing the number 4 and number 1 main bearing caps and bearings. If no damage is observed to these main bearing shells, further inspection is **not** necessary. On the products listed above, the number 4 main bearing journal is the first to receive lubricating oil. The number 1 main bearing journal is inspected to evaluate how far debris has spread through the lubricating system. Typically, more oil borne debris is found embedded in the main bearings than the rod bearings. The main bearings receive lubricating oil first and have a greater surface area to embed more of the debris. After inspection of the main bearings and crankshaft journals using the guidelines presented in this document, use the chart below to determine the appropriate repair direction.

Table 1, Repair Direction Following Main Bearing and Crankshaft Journal Inspection			
Main Bearings Reuse Criteria Met ¹ ?	Crankshaft Reuse Criteria Met ² ?	Repair Direction	Further Inspections
Yes	Yes	Re-install original components.	None.
No	Yes	Replace all main bearings with new.	Connecting rod bearings.
No	No	See Warranty Considerations Section Below.	

1. Use guidelines presented in Service Bulletin 3810387, Analysis and Prevention of Bearing Failures.
2. Use guidelines presented below in Table 2 of this bulletin.

Warranty Considerations

Only damage categories for non-catastrophic failure modes are shown in the Reuse Guidelines Section below. Any crankshaft exhibiting cracks, gross spalling/galling, or bearing seizure can **not** be reused.

If the crankshaft does **not** meet reuse guidelines:

- Dealers should contact RAPIDSERVE™ Catastrophic group for preauthorization if the engine is under warranty.
- Cummins® Distributors should follow their local catastrophic process or contact RAPIDSERVE™ Catastrophic group for repair direction if the engine is under warranty.

Reuse Guidelines

Objective analysis of crankshaft journal condition, as represented in Figures 1 through 7, can be made by following the reuse information presented in Table 2. The reuse criteria applies equally to the crankshaft main bearing journal and crankshaft connecting rod journal.

Table 2, Reuse Guidelines for Figures 1 through 6

Figure	Damage Category	Reuse
1	New/Normal Condition	Unconditional Reuse ¹
2	Scoring - Minor to Moderate	Unconditional Reuse ¹
3	Scoring - Heavy to Severe	Do Not Reuse
4	Frosted Bands	Unconditional Reuse ¹
5	Varnishing	Conditional Reuse ²
6	Raised Burr	Conditional Reuse ²

1. Unconditional Reuse – Do **not** replace crankshaft.

2. Conditional Reuse – Reference the description following each of the images depicting the type of damage.

Note : For all Figures below, scratches or scores appearing across the journal are **not** a result of damage during operation, but are typically a result of installation, disassembly, or handling practices.

New/Normal Condition

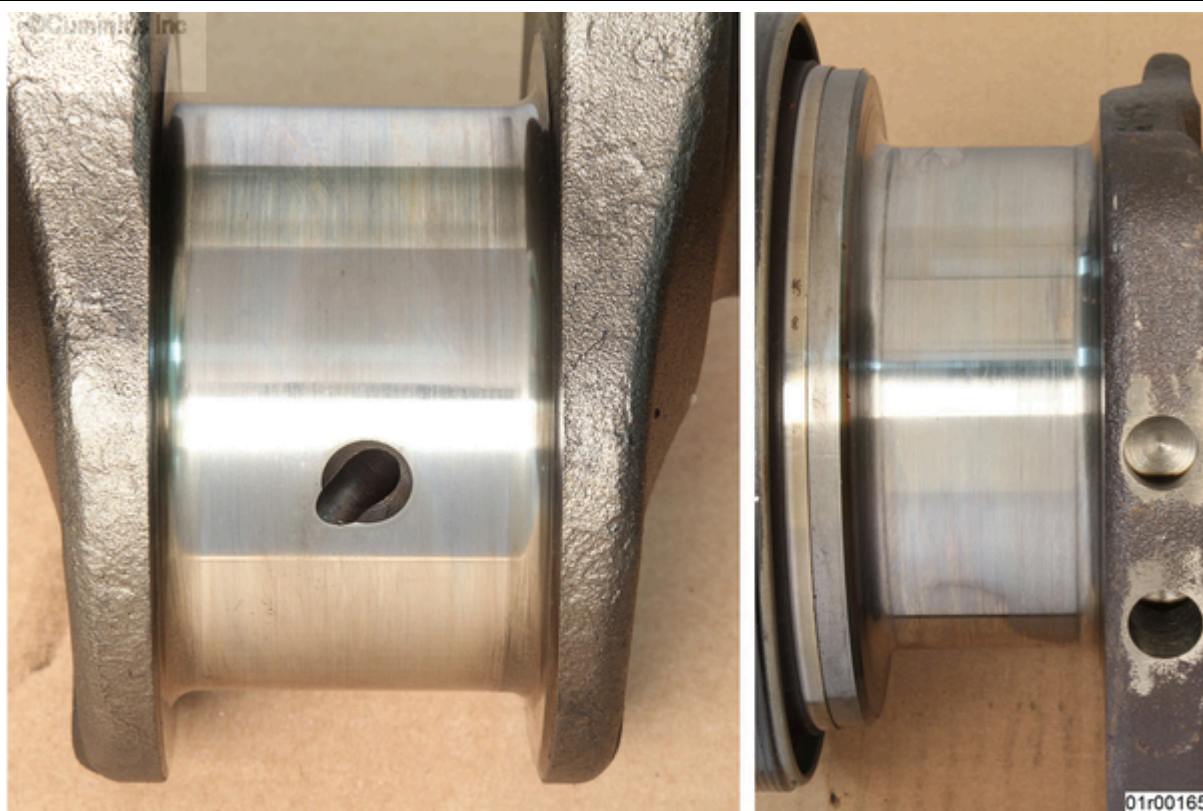


Figure 1, In-Service Rod and Main Journal – Normal Condition – Unconditional Reuse

Appearance

- Figure 1 is representative of undamaged crankshaft journals in operation. Note that oily residue observed is a result of cleaning and is **not** an indication of bluing or varnishing

depicted below.

Cause

- Normal.

Action

- Unconditional Reuse.

Scoring - Minor to Moderate

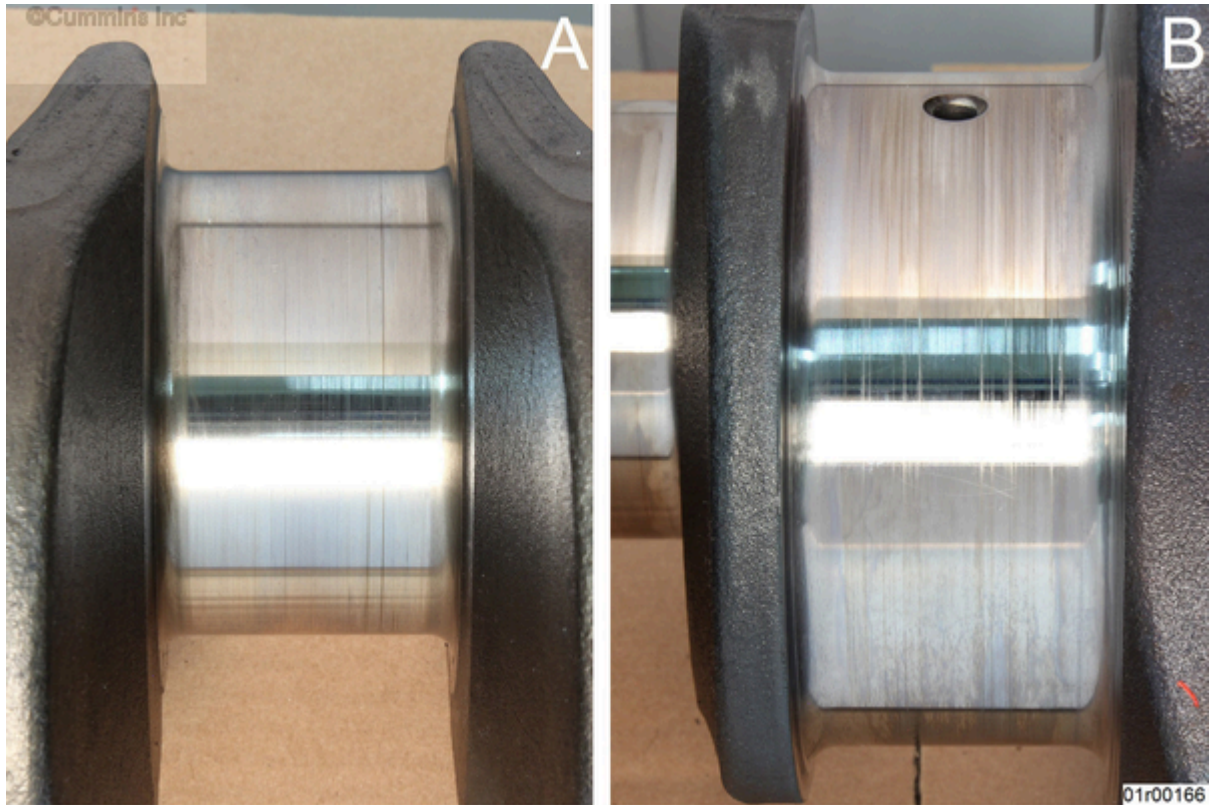


Figure 2, Journal Scoring – Minor to Moderate – Unconditional Reuse

Appearance

- Figure 2 shows traditional journal scoring. The scores shown can be felt with a fingernail, but just barely disrupt the surface material. Traverse motion with a fingernail could be interrupted slightly, but **not** with an abrupt 'catch of the fingernail'. These surface disruptions are minor and do **not** create a raised burr along the edge of the score.

Cause

- Debris in the lubricating oil.

Action

- Unconditional Reuse.
- The quantity and relative location of the scores are important. A small number of visible score marks away from the fillet radii and evenly distributed across the bearing surface are acceptable

Scoring - Heavy to Severe

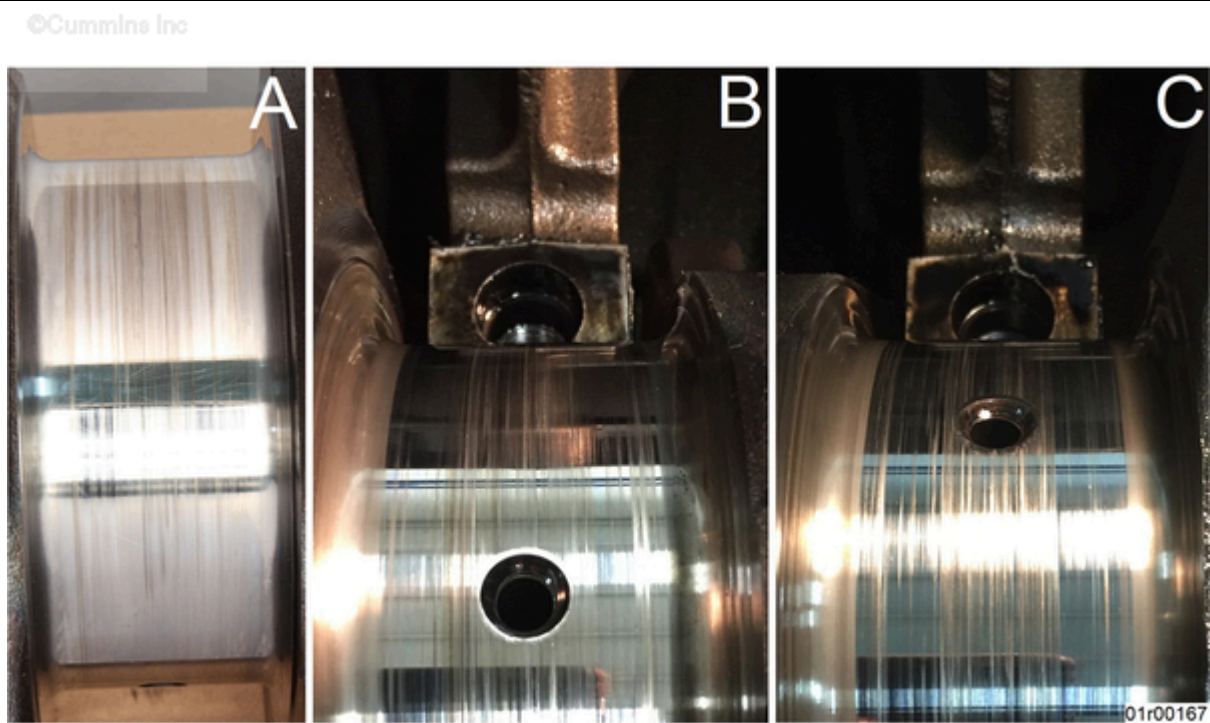


Figure 3, Journal Scoring – Heavy to Severe Damage – Do Not Reuse

Appearance

- The images in Figure 3 show greater degrees of journal scoring in increasing order of severity from left to right. These scores can be felt with a fingernail, but traverse motion will be frequently and abruptly interrupted. These scores are more numerous, and/or more concentrated. The depth of these scores is sufficient to raise a burr along the edge of the score.

Cause

- Debris in the lubricating oil.

Action

- Crankshafts with scored journals of this severity **must not** be reused.

Frosted Bands

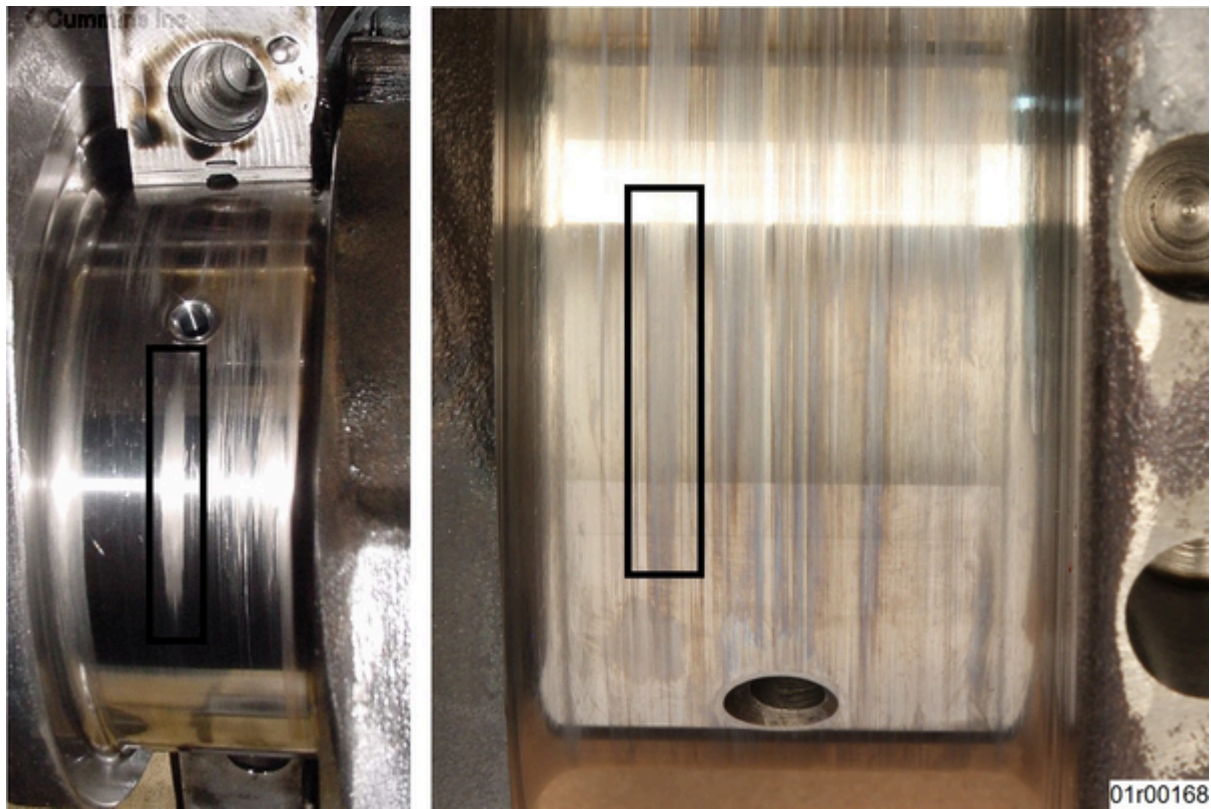


Figure 4, Frosted Bands – Unconditional Reuse

Appearance

- The images in Figure 4 depict frosted bands. Frosted bands can rarely be felt with a fingernail, and are primarily cosmetic.

Cause

- A common cause of frosted bands is very fine particle contamination in the lubricating oil, which does **not** result in scoring.

Action

- Unconditional Reuse.

Varnishing

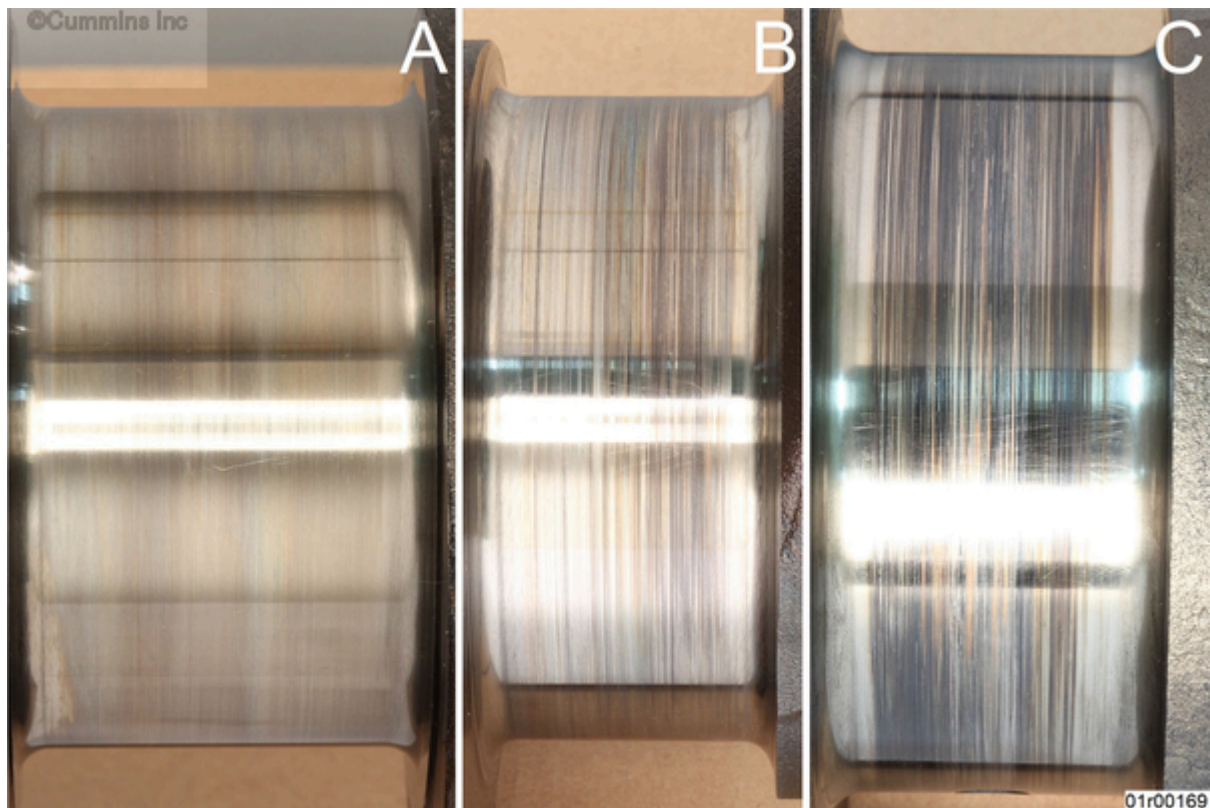


Figure 5, Journal Varnishing – A and B, Unconditional Reuse – C, Do Not Reuse

Appearance

- The images in Figure 5 show varying degrees of varnish build-up in increasing severity from left to right. Analyzing varnish build-up depends on a combination of coverage and color.
- Image A and B of Figure 5 represent mild to moderate varnish build-up with a tan to brown color and do **not** show defined witness marks at the edge of the bearing surface. Over time, this level of varnishing will clean from the surface with the normal operation of fresh engine lubricating oil.
- The varnish build-up shown in image C of Figure 4 is more progressed and may lead to reduced crankshaft running clearance. The color of the build-up in image C is purple to black and has witness marks that define the edge of the bearing surface indicated by the white lines.

Cause

- Varnish build-up is a result of lubricating oil oxidation due to excessive heat and/or extended oil drain intervals, among other less common sources of contamination.

Action

- The crankshafts from images A and B can be reused.
- The crankshaft from image C **must not** be reused.

Raised Burr

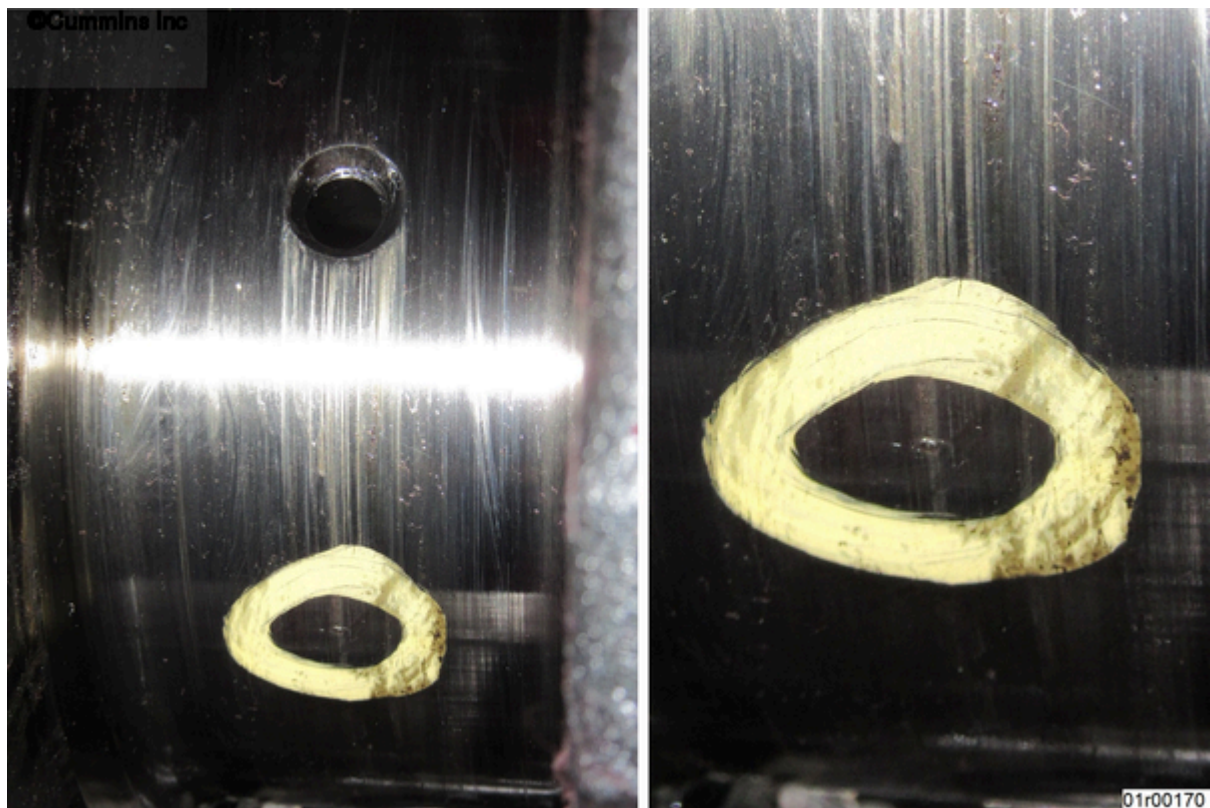


Figure 6, Raised Burr as a Result of Deep Scoring – Conditional Reuse

Appearance

- The images in Figure 6 show a burr remaining as a result of deep scoring. Burrs are typically observed at the end of a non-continuous score.

Cause

- Raised burrs is typically due to contaminated engine lubricating oil.

Action

- Attempt to remove the burr by lightly polishing the area with crocus cloth or emery paper. If successfully removed, the remainder of the damage **must** be assessed by the criteria shown for scoring in Figures 1 and 2. If the burr can **not** be removed after an attempt to polish, the crankshaft **must** be replaced.

Document History

Date	Details
2016-4-5	Module Created
2016-8-18	Updated Product Affected section.

